THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The probability of an event E always satisfies: 1

(a) $-1 \le P(E) \le 1$ **(b)** $P(E) < 0$ **(c)** $0 \le P(E) \le 1$ **(d)** $P(E) > 1$

2. A sequence in which each term is obtained by adding a fixed number to the previous term is called: 1

(a) an arithmetic progression **(b)** a Fibonacci sequence
(c) a geometric progression **(d)** a constant sequence

3. The radius of a circle is 25 cm and a chord is 30 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 20 **(b)** 21 **(c)** 19 **(d)** 22

4. The area of a triangle with sides 7 cm, 15 cm and 20 cm is (cm^2): 1

(a) 44 **(b)** 42 **(c)** 43 **(d)** 41

5. A die is thrown 100 times with frequencies of $1, \dots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 5 is: 1

(a) $\frac{4}{25}$ **(b)** $\frac{7}{51}$ **(c)** $\frac{7}{50}$ **(d)** $\frac{43}{50}$

6. If $t_n = 3n^2 - 4$, then t_4 is: 1

(a) 43 **(b)** 44 **(c)** 45 **(d)** 46

7. The chord AB subtends 88° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 54° **(b)** 44° **(c)** 64° **(d)** 34°

8. The semi-perimeter of a triangle with sides 9 cm, 10 cm and 17 cm is (cm): 1

(a) 19 **(b)** 20 **(c)** 18 **(d)** 17

9. In a survey of 400 families, 90 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{31}{40}$ **(b)** $\frac{9}{40}$ **(c)** $\frac{9}{41}$ **(d)** $\frac{1}{4}$

10. Which term of the sequence $t_n = 4n - 3$ is 81? 1

(a) 23th term **(b)** 21th term **(c)** 22th term **(d)** 20th term

11. The chord AB subtends 98° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 141° **(b)** 151° **(c)** 121° **(d)** 131°

12. The area of a circle with circumference 132 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 1387 **(b)** 1385 **(c)** 1388 **(d)** 1386

13. A letter is chosen at random from the word RAJASTHAN. The probability that it is a vowel is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

14. A sequence has $t_1 = 9$ and $t_{n+1} = t_n + 4$. Then t_6 equals: 1

(a) 31 **(b)** 30 **(c)** 28 **(d)** 29

15. In a cyclic quadrilateral $ABCD$, $\angle A = 68^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 78° **(b)** 58° **(c)** 68° **(d)** 88°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Three coins are tossed together. Find the probability of getting exactly one head. 2

17. Write the first five terms of the sequence with $t_1 = 1$ and $t_{n+1} = 2t_n + 1$. 2

18. In a circle of radius 26 cm, a chord is 24 cm from the centre. Find its length. 2

19. Find the area of a triangle with sides 7 cm, 15 cm, 20 cm. 2

20. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 18, 12, 20, 15, 19, 16. Find the experimental probability of getting an even number. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A bag contains 6 red, 2 black and 7 white balls. A ball is drawn at random. Find the probability that it is 3

(i) red,

(ii) not red,

(iii) black or white.

22. Find which term of the AP $90, 85, 80, 75, \ldots$ equals 25. 3

23. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 15)^\circ$ and $\angle C = (3x - 10)^\circ$. Find x and $\angle A$. 3

24. Find the area of the quadrilateral with sides 4, 8, 14, 22 cm that is inscribed in a circle. 3

OR

In a survey of 250 students, 100 take the bus, 75 cycle and the rest walk. Find the probability that a randomly chosen student 3

(i) walks,

(ii) does not take the bus.

25. Find the sum of the first 10 terms of the AP $25, 22, 19, 16, \ldots$. 3

OR

A chord of length 16 cm is 15 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two parallel chords of lengths 16 cm and 30 cm lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. **5**

OR

A circular track has inner radius 21 m and outer radius 28 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹4 per m^2. **5**

27. A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 18, 12, 20, 15, 19, 16. Find the experimental probability of **5**

(i) a prime number,

(ii) a number greater than 4,

(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A taxi charges a fixed ₹150 and ₹25 per km. **5**

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 18 km.

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND